Multi-Step ECG Forecasting Using Temporal Convolutional Networks with Grey Wolf Optimizer-Based Hyperparameter Tuning

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***Abstract*—Accurate multi-step forecasting of electrocardiogram (ECG) signals is critical for real-time cardiac monitoring and early anomaly detection. This paper investigates the application of Temporal Convolutional Networks (TCN) for multi-output ECG forecasting on the MIT-BIH Arrhythmia Database. We evaluate 21 patient records across four prediction horizons ($H \in \{1, 10, 15, 20\}$) using a per-patient training paradigm. We propose and compare three TCN configurations: (1) the original paper configuration with a single residual block and 256 filters, (2) our deeper architecture with three residual blocks and 64 filters, and (3) a Grey Wolf Optimizer (GWO)-tuned configuration that searches over filter count, dropout rate, and learning rate. Our deeper architecture achieves a mean R^2 of 0.8991 at H=10, outperforming the paper configuration (R^2=0.8776) while training 23% faster. The GWO-optimized model further improves to R^2=0.9020 at H=10 with 2.4% lower RMSE. Additionally, we conduct a preliminary inter-patient generalization study using Reversible Instance Normalization (RevIN), demonstrating a 6.8% RMSE improvement over standard TCN in the cross-patient setting. Future work directions include multi-objective optimization (MOGWO) balancing accuracy and computational cost, and integrating RevIN with the GWO-optimized architecture.**

***Index Terms*—Temporal Convolutional Network, ECG forecasting, Grey Wolf Optimizer, hyperparameter optimization, multi-step prediction, RevIN**

I. Introduction

Electrocardiogram (ECG) signal forecasting has emerged as a key enabler for predictive cardiac monitoring systems. Unlike classification tasks that identify arrhythmias post-hoc, forecasting enables proactive alerts by predicting future signal morphology from observed patterns [1]. Multi-step forecasting, where the model predicts H future time steps simultaneously, is particularly relevant for clinical decision support, as it provides a temporal window for intervention [2].

Temporal Convolutional Networks (TCN) have shown strong performance in sequence modeling tasks, rivaling or exceeding recurrent architectures such as LSTM and GRU [15]. TCNs leverage causal dilated convolutions to capture long-range temporal dependencies with parallelizable computation, making them well-suited for real-time physiological signal processing [3].

Despite the promise of TCNs for ECG applications, two critical challenges remain underexplored. First, the sensitivity of TCN performance to architectural hyperparameters—particularly filter count, number of residual blocks, dropout rate, and learning rate—necessitates systematic tuning beyond manual search [11]. Second, the inherent inter-patient variability in ECG morphology limits the generalizability of per-patient models, motivating normalization strategies that can bridge distributional shifts [16].

This paper makes the following contributions:

1) We reproduce and extend the baseline TCN configuration from Dudukcu et al. [3], demonstrating that a deeper, narrower architecture (3 blocks, 64 filters) outperforms the original (1 block, 256 filters) across all four prediction horizons while reducing training time.
2) We apply Grey Wolf Optimizer (GWO) [17] for automated hyperparameter tuning of the TCN, yielding further accuracy gains at H=10.
3) We present a preliminary study on Reversible Instance Normalization (RevIN) [16] for inter-patient ECG forecasting, showing measurable improvements in cross-patient generalization.
4) We outline future directions including multi-objective GWO (MOGWO) for jointly optimizing accuracy and training time, and integrating RevIN with the GWO-optimized architecture.

II. Related Work

A. Deep Learning for ECG Analysis

Deep learning approaches for ECG signals have evolved from classification-focused CNNs [4] to sequence-to-sequence forecasting models. Recurrent architectures including LSTM and GRU have been widely applied to ECG prediction [5], but suffer from sequential computation bottlenecks and vanishing gradient issues over long sequences. Barla et al. [4] combined CNN-LSTM-Transformer architectures with metaheuristic optimization for ECG classification, demonstrating the potential of hybrid deep learning and optimization approaches. Wang et al. [6] explored reservoir computing for ECG classification, highlighting the diversity of computational paradigms applied to cardiac signals.

B. Temporal Convolutional Networks

Bai et al. [15] demonstrated that TCNs can match or exceed recurrent networks on standard sequence benchmarks. The key architectural components—causal convolutions, exponentially increasing dilation rates, and residual connections—enable large receptive fields without sacrificing computational efficiency. Dudukcu et al. [3] proposed a hybrid TCN-RNN architecture for chaotic time series prediction, showing that TCN's ability to capture temporal patterns complements recurrent models. For physiological signal applications, TCNs offer deterministic inference time and straightforward parallelization on GPU hardware [14].

C. Metaheuristic Hyperparameter Optimization

Grey Wolf Optimizer (GWO), proposed by Mirjalili et al. [17], mimics the social hierarchy and hunting behavior of grey wolves. Massaoudi et al. [7] applied GWO to optimize TCN hyperparameters for power system transient stability assessment, demonstrating the effectiveness of metaheuristic tuning for temporal convolutional architectures. Recent work has extended GWO to multi-objective formulations (MOGWO) [18] for balancing competing objectives such as accuracy and model complexity.

D. Instance Normalization for Time Series

Kim et al. [16] proposed Reversible Instance Normalization (RevIN) to address distribution shift in time series forecasting. By normalizing each input instance by its own statistics and denormalizing the output, RevIN enables models to generalize across heterogeneous distributions without explicit domain adaptation. This is particularly relevant for ECG signals, where inter-patient variability in amplitude, baseline wander, and morphology creates significant distributional differences [8]. Bahrami and Fotouhi [8] investigated inter-patient, intra-patient, and patient-specific evaluation paradigms for ECG classification, showing that inter-patient settings remain the most challenging. Merdjanovska and Rashkovska [9] and Imtiaz and Khan [10] further studied cross-database generalization for arrhythmia detection, confirming that domain shift across patients and recording conditions remains a central obstacle.

III. Methodology

A. Dataset

We use the MIT-BIH Arrhythmia Database [20], a standard benchmark for ECG analysis containing 48 half-hour recordings sampled at 360 Hz. Following the protocol of [3], we select 21 patient records (excluding records 111 and 118 due to signal quality issues) and extract the MLII lead. The patient IDs used are: 100–109, 112–117, 119, 121–124.

For each patient, we extract the first 100,000 samples and partition them as:

- **Training set:** first 40,000 samples
- **Validation set:** next 10,000 samples
- **Test set:** remaining 50,000 samples

Each partition is normalized independently using Min-Max scaling fitted on the training set only, preventing information leakage. A lookback window of L=10 time steps is used to construct input–output pairs, where the model receives L past values and predicts the next H values simultaneously (multi-output strategy).

B. TCN Architecture

Our TCN follows the standard architecture with stacked residual blocks. Each residual block consists of two causal dilated convolution layers, each followed by batch normalization, ReLU activation, and dropout. The dilation rate doubles with each block ($d_i = 2^i$ for block i), providing an exponentially growing receptive field. A $1{\times}1$ convolution is applied to the skip connection when the input and output channel dimensions differ.

The network takes an input tensor of shape $(B, L, 1)$ where B is the batch size and L=10 is the lookback window. After the final residual block, we extract the last time step's representation and pass it through a dense layer with ReLU activation, followed by a linear output layer of size H for multi-step prediction.

The receptive field of the TCN is given by:

$$R = 1 + 2 \cdot (k-1) \cdot \sum_{i=0}^{n-1} 2^i = 1 + 2(k-1)(2^n - 1) \quad (1)$$

where k is the kernel size and n is the number of residual blocks. For our configuration (k=3, n=3), $R = 1{+}2{\cdot}2{\cdot}7 = 29$, which exceeds the lookback window of 10 and ensures full coverage of the input sequence.

C. Configuration Comparison

We evaluate three TCN configurations, summarized in Table I.

TABLE I: TCN Configuration Comparison

Parameter	Paper	Ours	GWO
Residual blocks	1	3	4
Filters	256	64	256
Kernel size	3	3	3
Dropout	0.1	0.1	0.109
Batch size	128	256	128
Learning rate	0.001	0.001	0.00207
Epochs	200	200	200
Patience	50	50	50
Receptive field	5	29	61

The **Paper** configuration reproduces the settings reported in [3]: a single residual block with 256 filters. The **Ours** configuration uses a deeper network (3 blocks) with fewer filters (64), increasing the receptive field from 5 to 29 while reducing the parameter count. The **GWO** configuration is obtained through automated hyperparameter optimization as described in Section III-D.

D. Grey Wolf Optimizer for Hyperparameter Tuning

We employ the Grey Wolf Optimizer [17] implemented via the MealPy library [19] to tune three hyperparameters while keeping others fixed (n_{blocks}=4, k=3, batch size=128):

- **Filter count** $f \in \{32, 64, 128, 256\}$: encoded as an integer index 0–3
- **Dropout rate** $d \in [0.05, 0.40]$: continuous
- **Learning rate** $\eta \in [10^{-4}, 10^{-2}]$: continuous in log-space

The objective function trains a TCN on 5 pilot patients (records 100, 103, 109, 117, 124) at H=10 with a reduced budget of 100 epochs and patience of 20. The fitness value is the mean RMSE across the pilot patients' test sets. The GWO search uses a population of 8 wolves over 4 epochs, totaling 32 evaluations.

The best solution was found at evaluation 5 with mean pilot RMSE of 0.02615, yielding: f=256, d=0.109, η=0.00207. This configuration was then evaluated on all 21 patients at H=10 with full training budget (200 epochs, patience 50).

E. Reversible Instance Normalization (RevIN)

To explore inter-patient generalization, we implement RevIN [16] as a wrapper around the TCN. RevIN normalizes each input window independently:

$$\hat{x} = \gamma \cdot \frac{x - \mu_x}{\sqrt{\sigma_x^2 + \epsilon}} + \beta \tag{2}$$

where μ_x and σ_x^2 are the per-instance mean and variance, and γ, β are learnable affine parameters. The output is denormalized by reversing the transformation:

$$y = \sqrt{\sigma_x^2 + \epsilon} \cdot (\hat{y} - \beta)/\gamma + \mu_x \tag{3}$$

For the inter-patient experiment, we train on 15 patients' combined data and test on 6 held-out patients (records 117, 119, 121, 122, 123, 124) at H=10, comparing TCN with and without RevIN. Both models use the baseline "Ours" configuration.

F. Evaluation Metrics

We evaluate performance using three metrics computed on the original (inverse-scaled) signal:

- **RMSE**: Root Mean Square Error, measuring average prediction magnitude error
- **MAE**: Mean Absolute Error, providing a robust average error estimate
- R^2: Coefficient of determination, indicating the proportion of variance explained

All metrics are computed on the flattened multi-step predictions across all test samples for each patient.

IV. Experimental Results

All experiments were conducted on NVIDIA GPU hardware using TensorFlow/Keras. Each configuration was evaluated on all 21 patients independently, with per-patient models trained from scratch.

A. Ours vs. Paper Configuration

Table II presents the mean results across all 21 patients for each horizon and configuration.

TABLE II: Mean Results Across 21 Patients (Ours vs. Paper Configuration)

H	Config	RMSE	MAE	R^2	Time (s)
1	Paper	0.0061	0.0037	0.9952	226.6
	Ours	0.0052	0.0036	0.9970	169.8
10	Paper	0.0344	0.0148	0.8776	207.7
	Ours	0.0304	0.0134	0.8991	159.5
15	Paper	0.0471	0.0200	0.7847	202.0
	Ours	0.0433	0.0185	0.8108	160.4
20	Paper	0.0571	0.0253	0.6894	188.6
	Ours	0.0539	0.0236	0.7157	156.4

Our configuration consistently outperforms the paper configuration across all horizons and metrics. At H=10, our model achieves 11.6% lower RMSE (0.0304 vs. 0.0344), 9.4% lower MAE, and 2.4% higher R^2, while training 23.2% faster on average. The deeper architecture with fewer filters provides a better accuracy–efficiency trade-off: the increased receptive field (29 vs. 5 time steps) captures longer-range dependencies, while the reduced filter count lowers computational cost.

As expected, performance degrades with increasing horizon for both configurations. However, the gap widens at longer horizons: our configuration's R^2 advantage grows from +0.18% at H=1 to +3.8% at H=20, suggesting that the larger receptive field becomes increasingly important for multi-step forecasting.

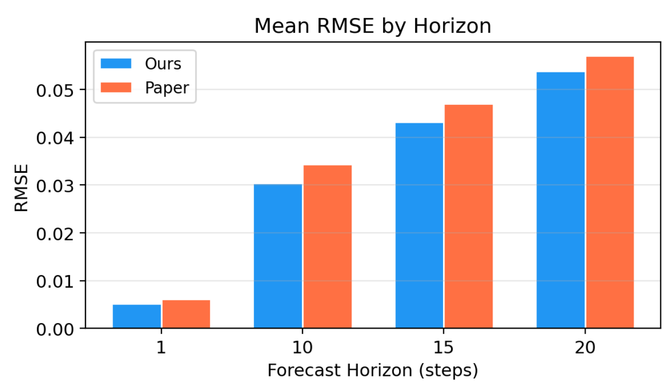

Fig. 1: Mean RMSE comparison across horizons: Ours vs. Paper configuration.

B. GWO-Optimized Results at H=10

Table III compares the GWO-optimized configuration against our baseline at H=10.

The GWO-optimized model improves RMSE by 2.4% and MAE by 3.8% over our baseline, pushing mean R^2 above 0.90. However, the optimized configuration (256 filters, 4 blocks) incurs a 3.7× increase in training time per patient

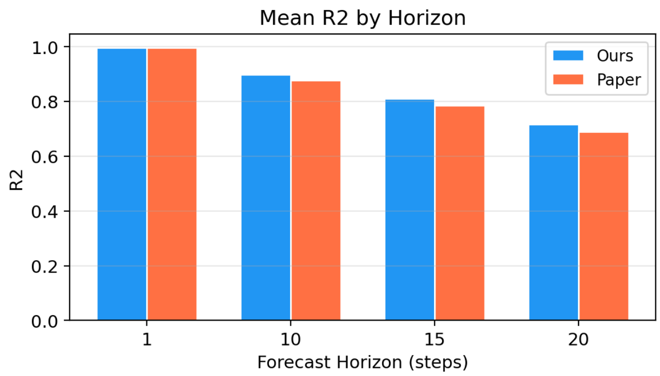

Fig. 2: Mean R^2 comparison across horizons: Ours vs. Paper configuration.

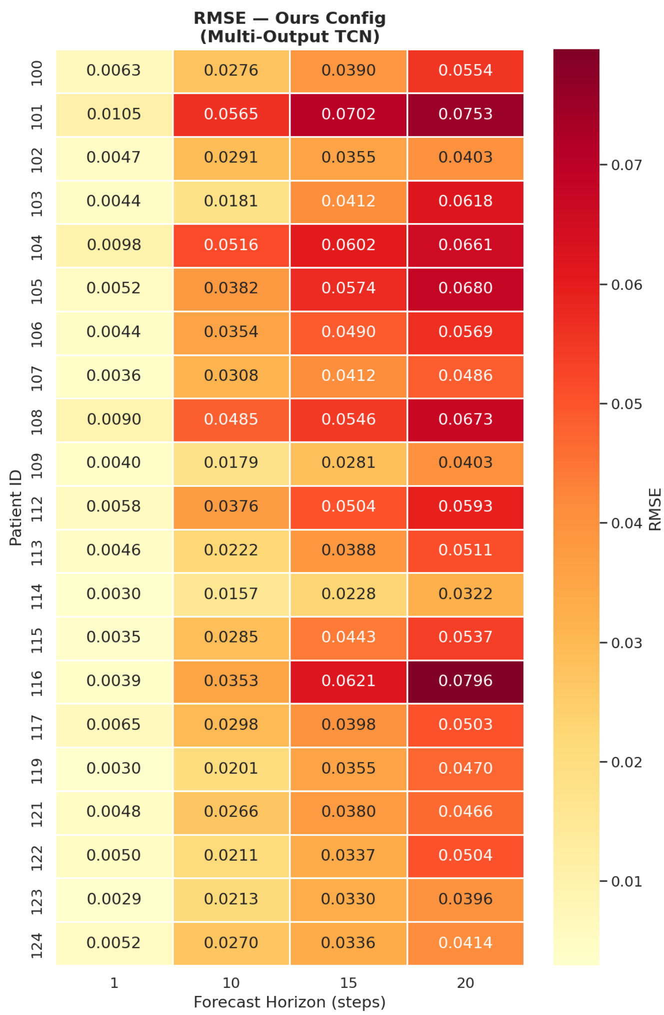

Fig. 3: Per-patient RMSE heatmap across horizons (Ours configuration).

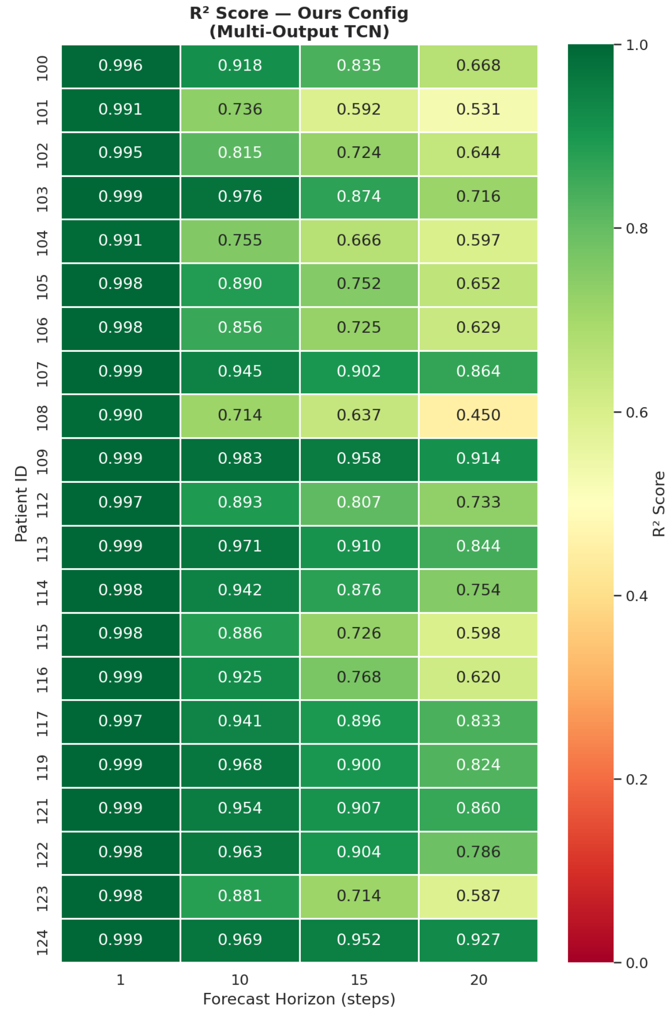

Fig. 4: Per-patient R^2 heatmap across horizons (Ours configuration).

TABLE III: GWO-Optimized vs. Ours Configuration at H=10 (Mean over 21 Patients)

Metric	Ours	GWO	Δ%
RMSE	0.0304	0.0297	−2.36%
MAE	0.0134	0.0129	−3.79%
R^2	0.8991	0.9020	+0.33%
Time (s)	159.5	591.5	+271%

due to the larger model capacity. This accuracy–cost trade-off motivates the multi-objective optimization approach discussed in Section V.

The GWO search converged quickly, finding the best solution at evaluation 5 out of 32 total evaluations. The optimal configuration favored high filter count (256) and moderate dropout (0.109), with a learning rate approximately twice the baseline (0.00207 vs. 0.001). Out of 21 patients, 20 showed improvement under the GWO configuration, with the largest gains observed for patients with high signal variability (e.g., Patient 114: −17.6% RMSE, Patient 109: −12.8% RMSE). The sole exception was Patient 101, where RMSE increased by 13.1%.

C. Prediction Quality

Fig. 5 and Fig. 6 show representative prediction plots for Patient 103, which achieved the highest R^2 values. At H=1, the TCN accurately tracks rapid signal transitions including QRS complexes. At H=10, the model captures the overall signal envelope with graceful degradation at sharp transitions.

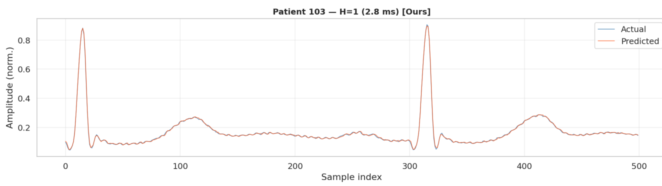

Fig. 5: Prediction vs. ground truth for Patient 103 at H=1 (Ours configuration, R^2=0.9986).

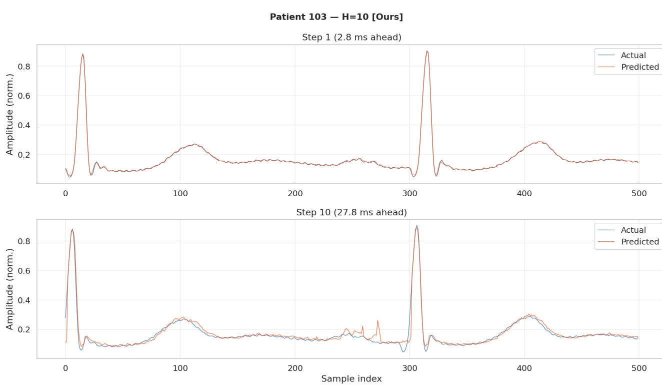

Fig. 6: Prediction vs. ground truth for Patient 103 at H=10 (Ours configuration, R^2=0.9758).

D. Per-Step Error Analysis

Fig. 7 shows how prediction error evolves across individual forecast steps. RMSE increases monotonically with step index, confirming the expected error accumulation in multi-step prediction. The rate of increase is steeper for the first 5 steps and plateaus toward the end of the horizon, suggesting that the model learns a smoothed representation of distant future values.

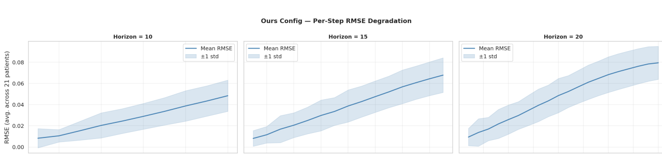

Fig. 7: Per-step RMSE across forecast horizon (Ours configuration, averaged over patients).

E. Inter-Patient Generalization with RevIN

Table IV presents the inter-patient experiment results, where models trained on 15 patients are tested on 6 unseen patients at H=10.

TABLE IV: Inter-Patient Results: TCN with and without RevIN (H=10)

Model	RMSE	MAE	R^2	Time (s)
TCN (no RevIN)	0.1870	0.0760	0.7318	650.5
TCN + RevIN	0.1742	0.0682	0.7585	642.4
Improvement	**6.8%**	**10.3%**	**+0.027**	—

RevIN provides consistent improvement across all metrics with negligible computational overhead (only 2 additional learnable parameters). The per-instance normalization effectively mitigates the distributional shift between patients' ECG amplitudes and baselines. Notably, the absolute performance is substantially lower than per-patient models (R^2=0.76 vs. 0.90), reflecting the fundamental challenge of inter-patient ECG generalization where morphological patterns vary significantly across individuals.

F. Training Dynamics

Fig. 8 illustrates representative training loss curves. Early stopping with patience of 50 epochs effectively prevents overfitting, with most models converging between 100–200 epochs. The validation loss plateau typically precedes the training loss plateau, confirming appropriate regularization through dropout and early stopping.

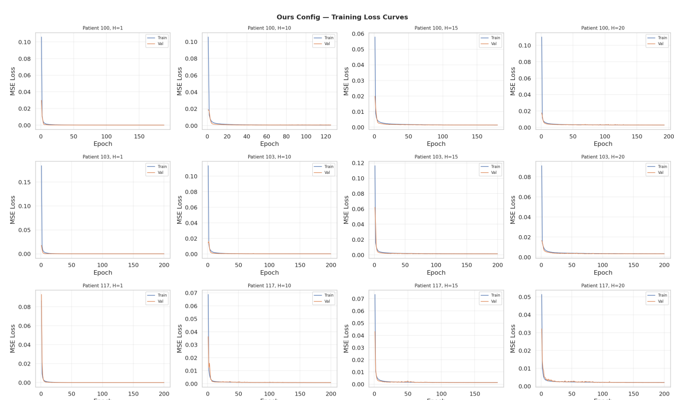

Fig. 8: Training and validation loss curves for selected patients (Ours configuration).

V. Future Work

Several promising directions extend from our current findings:

A. Multi-Objective GWO (MOGWO)

The GWO-optimized configuration improved accuracy by 2.4% RMSE but increased training time by 3.7×. To address this accuracy–efficiency trade-off, we plan to apply Multi-Objective Grey Wolf Optimizer (MOGWO) [18], which simultaneously optimizes two objectives:

$$\min_{\theta} [\mathrm{RMSE}(\theta),\ T_{\mathrm{train}}(\theta)] \tag{4}$$

where θ represents the hyperparameter vector and T_{train} is the total training time. MOGWO produces a Pareto front of non-dominated solutions, allowing practitioners to select configurations based on their computational budget. We also plan to explore literature on which specific hyperparameters yield the highest sensitivity for TCN-based time series models, enabling a more targeted search space definition.

B. RevIN Integration with GWO-Optimized Architecture

Our RevIN experiment used the baseline "Ours" configuration in an inter-patient setting. A natural extension is to combine RevIN with the GWO-optimized architecture and evaluate in both per-patient and inter-patient settings. The hypothesis is that RevIN's per-instance normalization can complement the optimized model's capacity, particularly for patients with high signal variability where the GWO model showed the largest gains.

C. Extended Hyperparameter Search

The current GWO search optimizes three parameters with two fixed (n_{blocks}, k). Future work will expand the search space to include:

- Number of residual blocks $n \in \{2, 3, 4, 5, 6\}$
- Kernel size $k \in \{2, 3, 5, 7\}$
- Batch size $b \in \{64, 128, 256, 512\}$

We will draw on existing hyperparameter sensitivity studies for temporal convolution architectures [11] to prioritize the most impactful parameters.

D. Multi-Horizon GWO Optimization

The current GWO search targets $H{=}10$ only. We plan to either (a) optimize separately per horizon, or (b) define a composite objective averaged across $H \in \{1, 10, 15, 20\}$ to find a single configuration that performs well across all prediction windows.

VI. Conclusion

We presented a systematic study of TCN architectures for multi-step ECG forecasting on the MIT-BIH Arrhythmia Database. Our deeper, narrower TCN configuration (3 blocks, 64 filters) outperformed the original paper configuration (1 block, 256 filters) across all four prediction horizons, achieving a mean R^2 improvement of 2.4% at $H{=}10$ with 23% faster training. Grey Wolf Optimizer-based hyperparameter tuning further improved accuracy, reaching $R^2{=}0.9020$ at $H{=}10$, though at a $3.7\times$ computational cost. A preliminary RevIN experiment demonstrated 6.8% RMSE improvement in inter-patient generalization, motivating further integration of normalization strategies with optimized architectures. Future work on MOGWO will address the accuracy–efficiency Pareto front to make automated tuning practical for deployment scenarios.

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